AI/ML for Chip Design and EDA

- Current Status and Future Perspectives

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Biography

- 최 규명
  - 1985.01 ~ 2004.12 … 삼성전자 반도체 연구원 ~ 수석연구원, CAE팀
  - 2005.01 ~ 2010.12 … 삼성전자 상무, Design Technology팀장
  - 2011.01 ~ 2013.12 … 삼성전자 전무, 기반설계센터장
  - 2014.01 ~ 2015.12 … 삼성전자 전무, LSI개발실장
  - 2016.03 ~ 2020.02 … 서울대학교 전기정보공학부 객원교수
  - 2019.02 ~ 2020.01 … Synopsys Inc., Senior Advisor
  - 2020.12 ~ 2024.12 … 서울대학교 전기정보공학부 산학협력중점교수 및 반도체공동연구소/공학전문대학원 책임연구원
  - 2025.01 ~ … 성균관대학교 반도체융합공학과 산학교수

  - 1983.02, 1985.02 … 한양대학교 전자공학과 학사, 석사
  - 1991.09 ~ 1995.10 … Univ. of Pittsburgh ECE Dept. 박사

MWC(Mobile World Congress) 2015

사그라다 파밀리아 성당(1882 착공, 2026 완공예정, 안토니오 가우디)

Contents

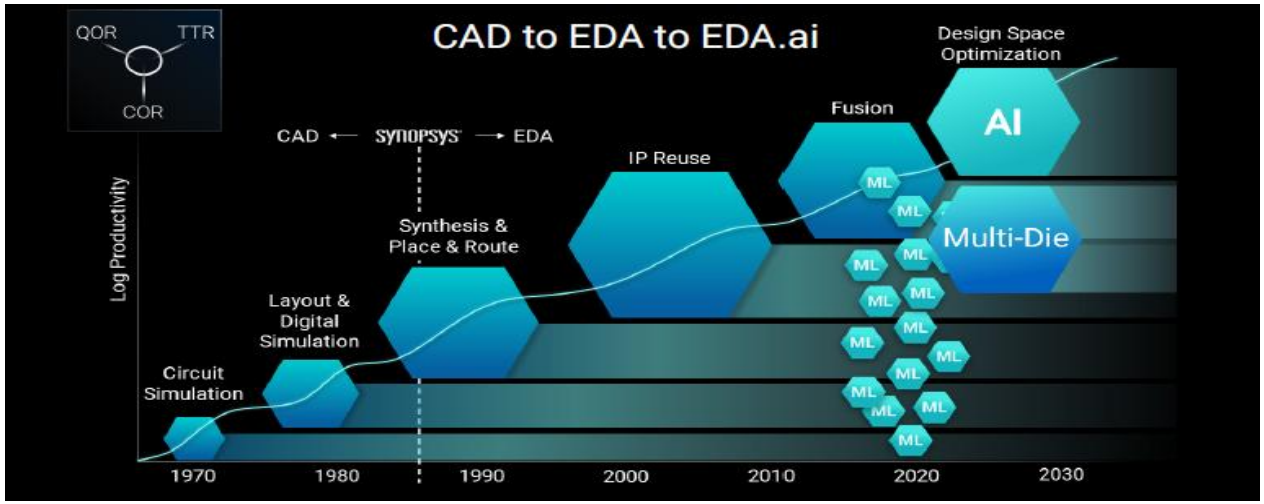
- **Technology Necessity and Current Status**
- Research Status at Academia
- Development and Adoption Status at Industry
- Generative AI for Chip Design

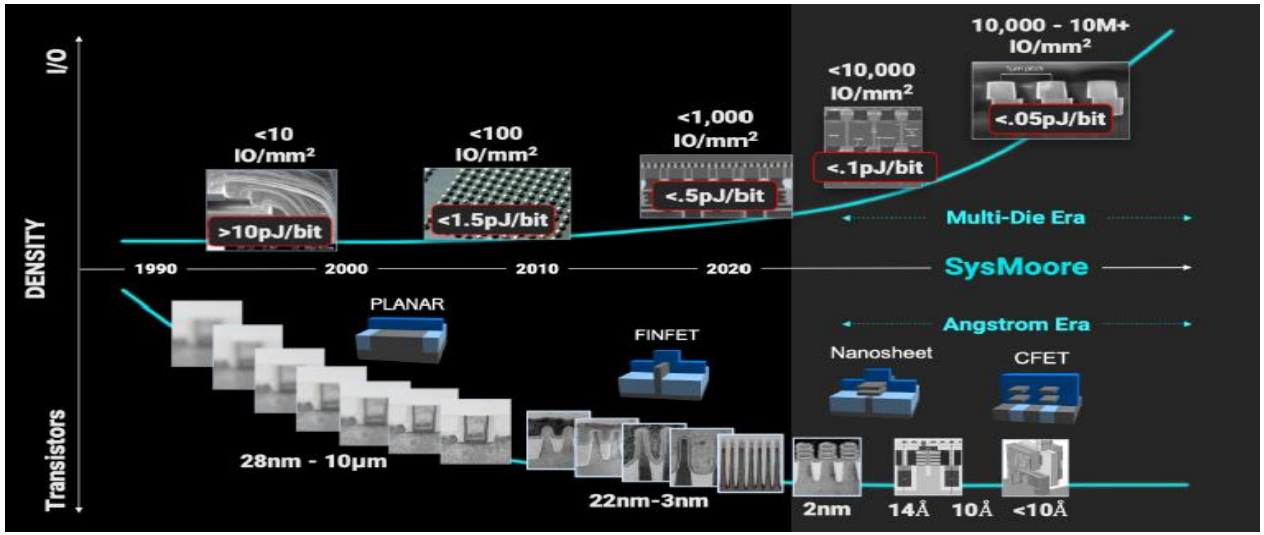
Semiconductor Design: CAD, EDA, AI/ML, Multi-Die

- SysMoore era: Systemic Complexity with a Moore's law ambition!

Computer-Aided Design to

Electronic Design Automation





Multi-Die systems:

Why Now?

* Source: Aart de Geus, CEO of Synopsys, "How quickly will multi-die systems change semiconductor design", Industry insight, Synopsys, 2023

Semiconductor Design Challenge

- Semiconductor design challenge
  - Process/Device innovation
    - Transistor … Planar → FinFET → NanoSheet → CFET
    - Lithography … RET advancement, ArF → EUV
  - IC Integration (More Moore + More than Moore)
    - 2D scaling → 3D integration (heterogeneous integration, package technology)
  - Diverse technology drivers (CPU → AP → AP, 5G, AI, Automotive, IoT, AR/VR)
    - AI chip design … AI hardware accelerator implementation
    - Memory-logic integration … PIM/PNM (Processor in/near Memory)
    - Different architectures … Different spec. requirement
  - Getting worse of design productivity and design cost
    - Design objectives … Performance, Power, Area → + Cost, Yield

Resolution of Semiconductor Design Challenge

- *Current EDA tools, mainly based on algorithmic approaches, have not resolved "Semiconductor Design Challenge" efficiently.*
- **Differentiation by Machine learning(ML)**
  - Integrated analysis of multi-objects by multi-sources
  - Accurate early prediction of design metrics (PPA, etc.)
  - Accurate but fast simulation/analysis by surrogate model
  - Fast design space exploration and optimization(DSE/DSO)
  - Increase automation coverage by sequential design optimization (analog, memory, etc.)

  → Reduced human resources by extreme automation
- **Resolution of semiconductor design challenge**

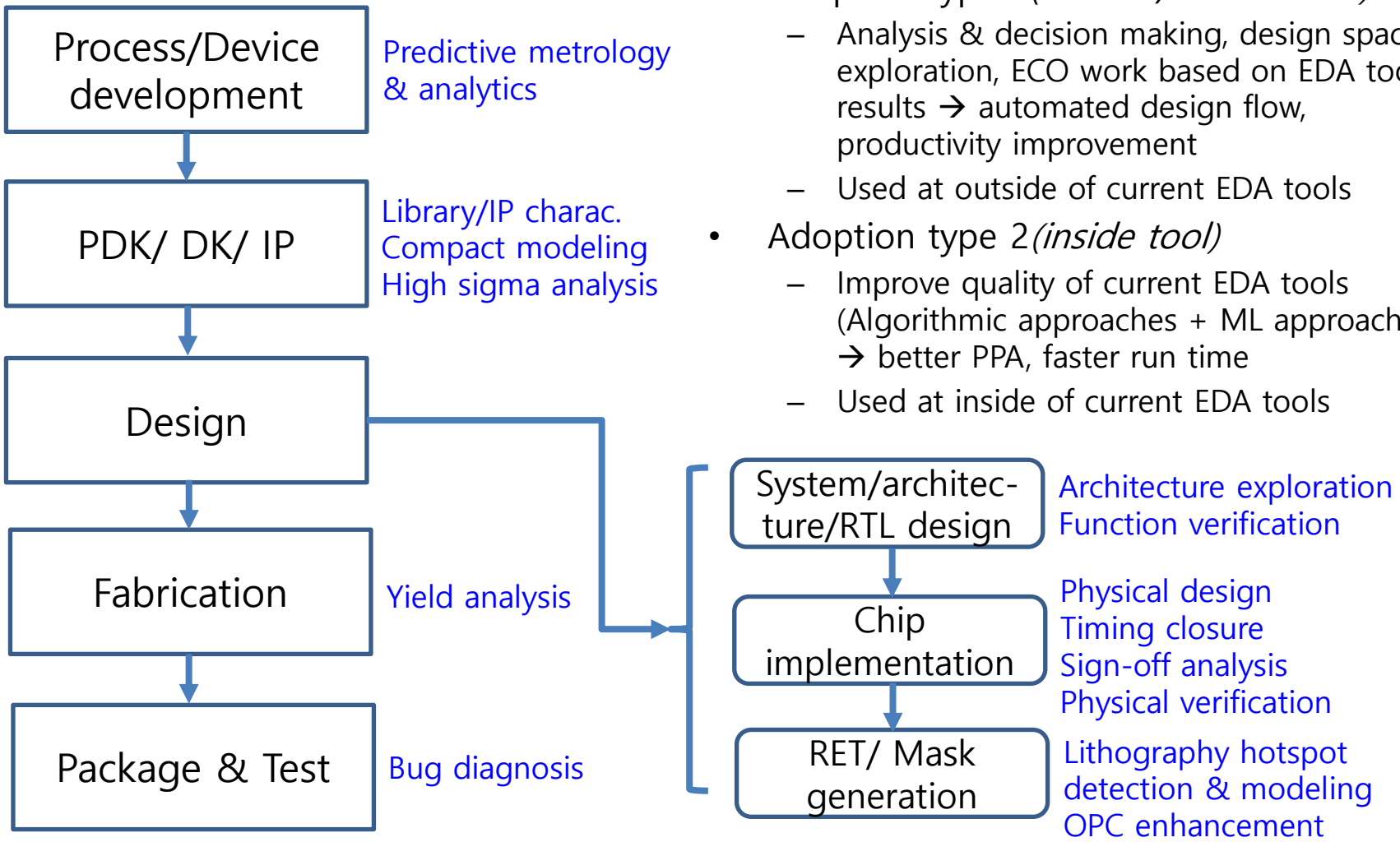
New Machine Learning Solution

Current EDA tool chain (algorithmic approaches) + Machine Learning (Data driven approaches)

Dig for current EDA Solution

Need more closer collaboration between foundries, design companies, EDAs

Machine Learning for EDA

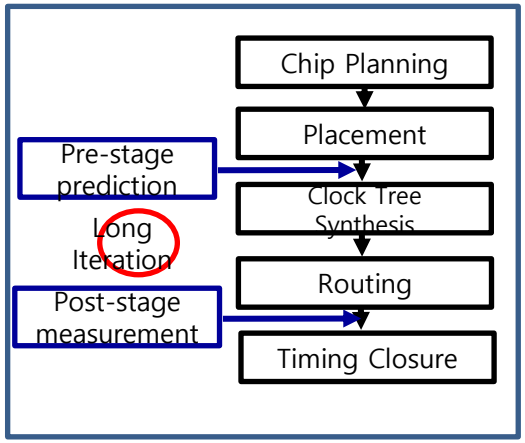


- Adoption type 1 *(outside/around tool)*
  - Analysis & decision making, design space exploration, ECO work based on EDA tools results → automated design flow, productivity improvement
  - Used at outside of current EDA tools
- Adoption type 2 *(inside tool)*
  - Improve quality of current EDA tools (Algorithmic approaches + ML approaches) → better PPA, faster run time
  - Used at inside of current EDA tools

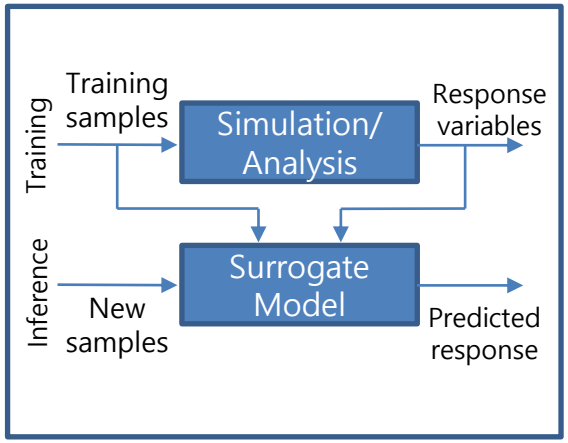
Machine Learning for EDA

- Areas
  - Design: System-level design, Verification, Chip implementation, RET/mask generation
  - Process/device development, Fabrication, Package & Test, etc.

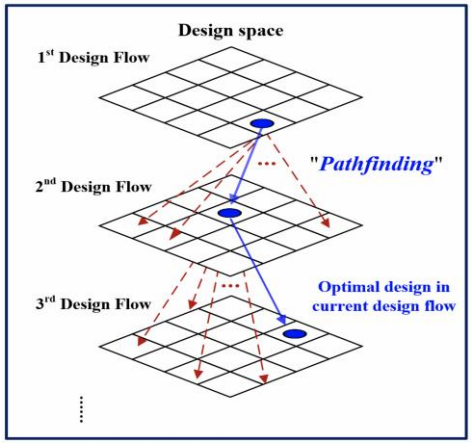
- Approaches
  - Supervised learning → Analysis, Prediction
  - Surrogate modeling → Simulation/analysis speed-up
  - Reinforcement learning → DSE/DSO, Sequential design optimization
  - Generative AI (Transformer, LLM: Large Language Model)



Early stage prediction



Simulation speed-up



Design space exploration/optimization

Research & Development Status

- Activities for Research and development have arisen since 2016 at major universities, EDA companies, and semiconductor companies.

- However, it is still early stage to be adopted in real designs.

- **USA**: Resurgence of technology development by national funds disappeared since end of 1990.
  - CAEML(NFS: 2017.1~), MAGESTIC(DARPA: 2018.7~), OpenROAD(DARPA: 2018.6~)
  - Group research between major universities mainly, including some companies
  - EDA and semiconductor companies join as consortium

- **China**: Connected research between China and Chinese professors in America, Hong Kong, Taiwan
  - Active open-source collaborative projects(iEDA, iPD)*
  - Show most active research activities in the world

- **Korea**: AI-EDA(nationally funded group research) project (2021.4~2024.12)

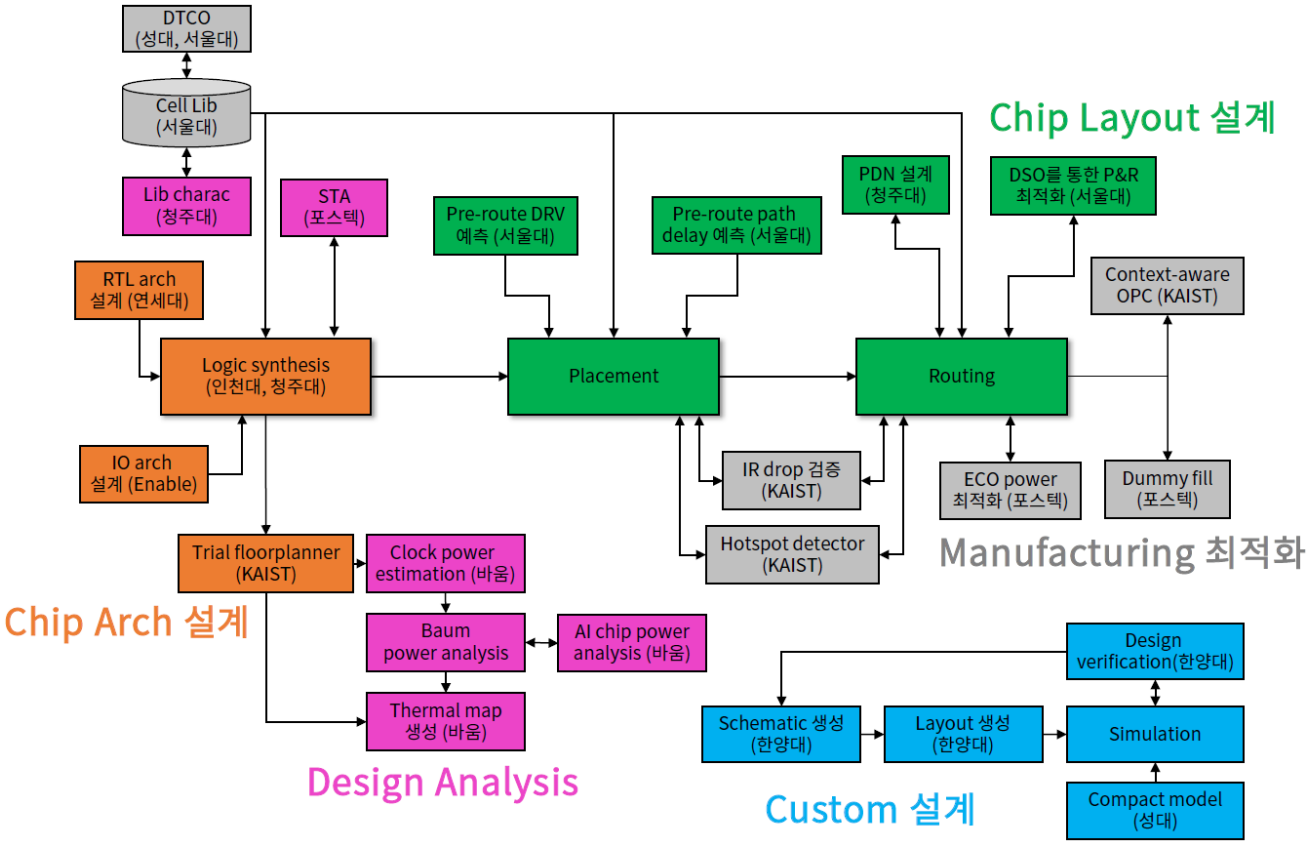
* iEDA: An open-source infrastructure of EDA, iPD: An open-source intelligent physical design toolchain
… Both of them was announced at ASPDAC-2024

AI-EDA - IITP Project: Software System for AI Semiconductor Design

- 주관 기관: KAIST
- 참여 기관:
  - 대학 … KAIST, 서울대, 성균관대, 연세대, 인천대, 청주대, 포항공대, 한양대
  - 국내 EDA 기업

  … Baum, EnableDesign
- 과제 기간:

  2021.4.1~2024.12.31



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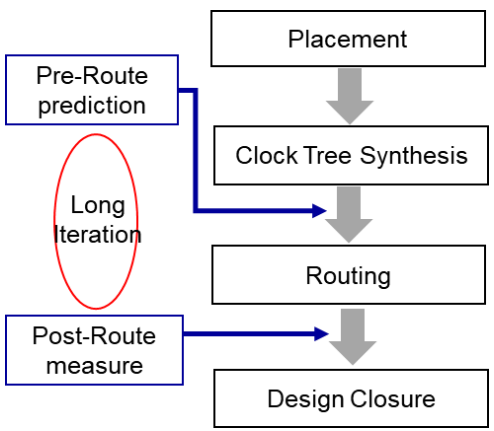
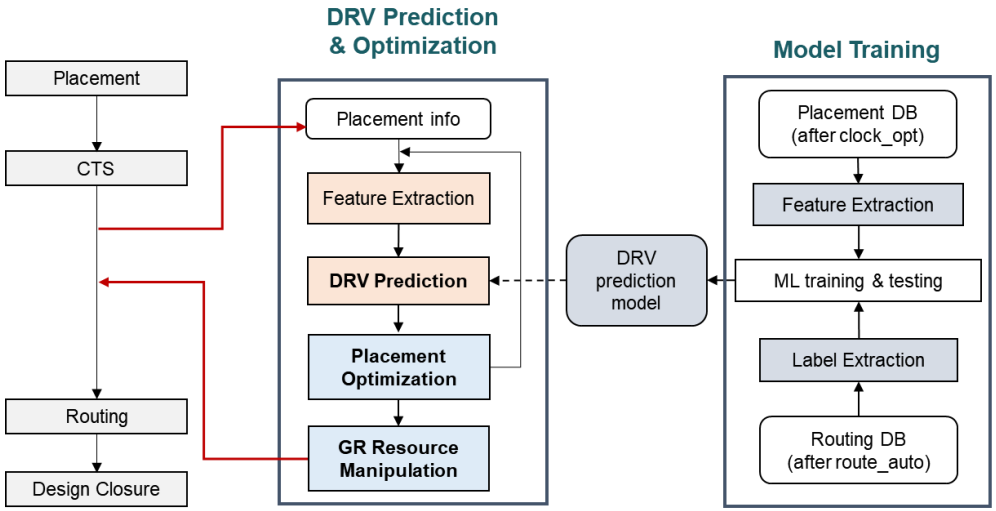
Pre-route DRV Prediction and Optimization[1,2,3]

- Motivation … Accurate prediction of DRV(design rule violation) and optimization to reduce DRV at pre-route stage → design time reduction
- Need to consider → Global Routing Congestion + Pin Accessibility
- Training ML-based DRC hotspot prediction model … supervised learning using GNN and U-net
- Predicting DRC hotspot during the pre-route stage of the P&R flow
- Placement optimization & Controlling global routing resources for DRV resolution



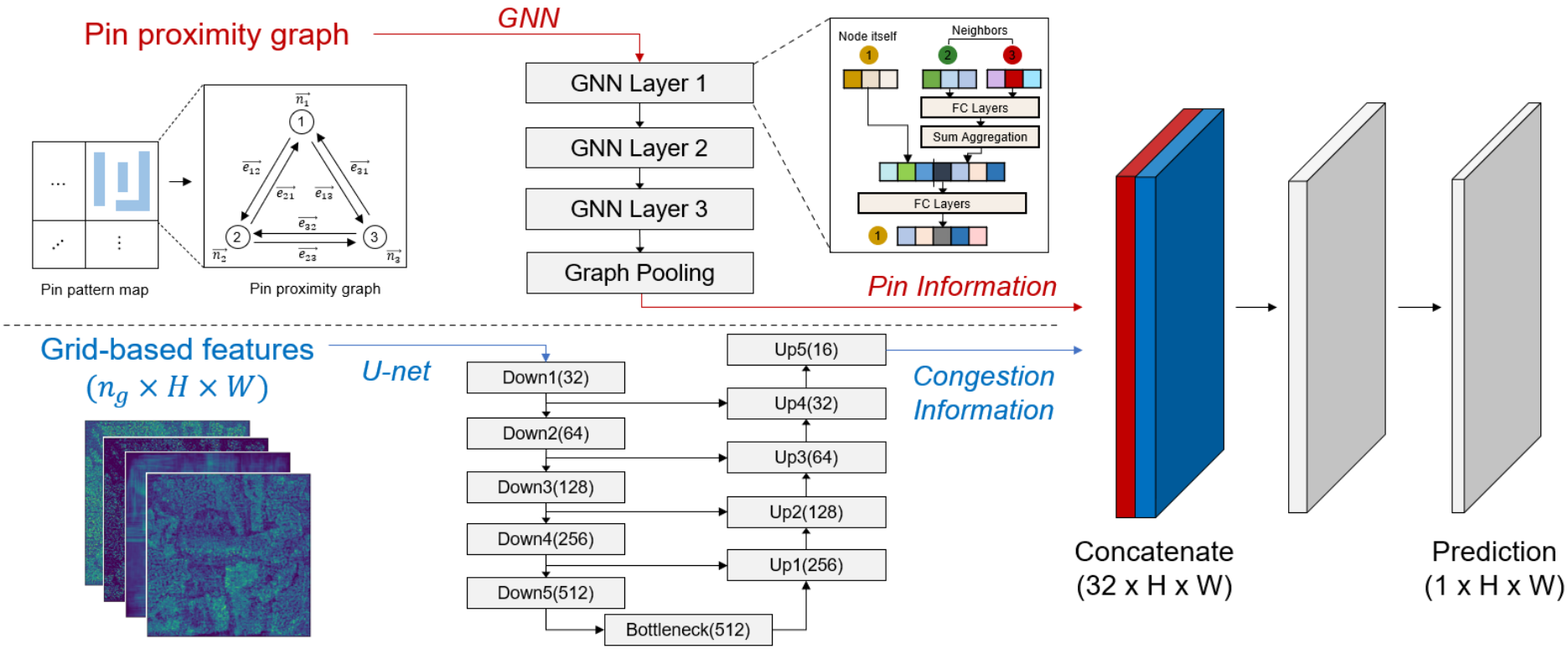
Motivation



Flowchart of proposed framework.

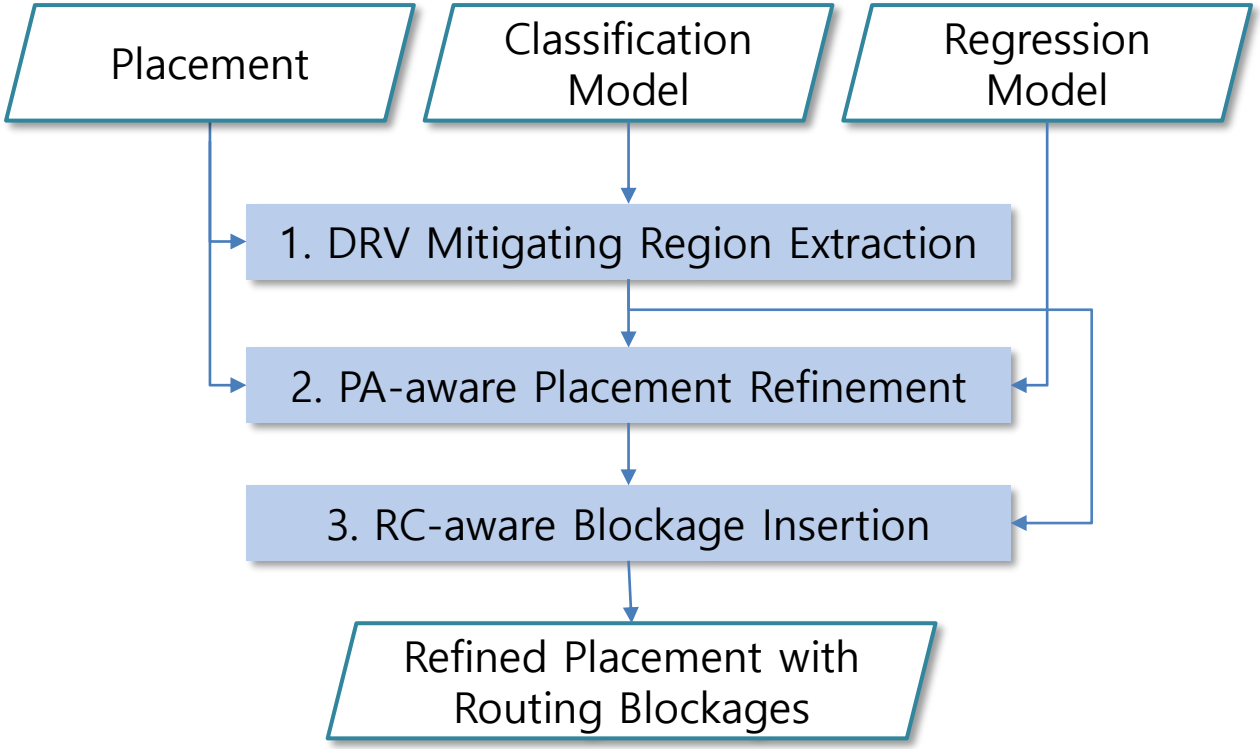
Pre-route DRV Prediction and Optimization

- DRV Prediction - Overall Model Architecture
  - Combined model of GNN(Graph Neural Network) and U-Net
    - GNN extracts pin information from pin proximity graph
    - U-net extracts congestion information from grid-based features
    - Producing DRC hotspot prediction from concatenated information



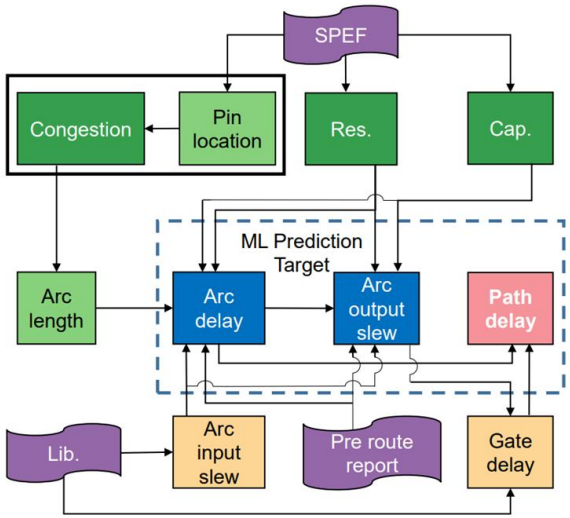
Pre-route DRV Prediction and Optimization

- DRV Optimization - Overview of DRV Resolution Methodology
  - Two types of DRV resolution strategies are proposed for each type of DRV.
    - Pin accessibility(PA)-related DRVs – Placement refinement
    - Routing congestion(RC)-related DRVs – Routing resource manipulation

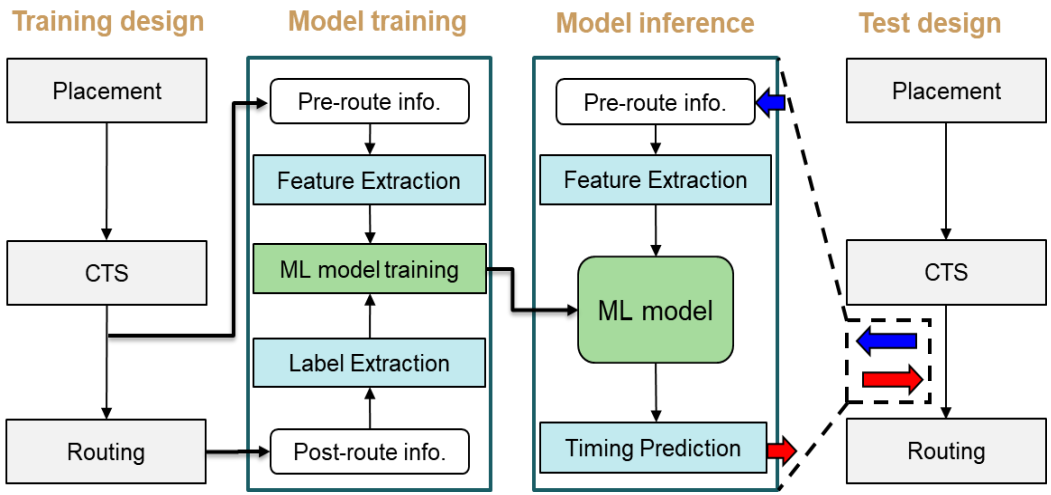


Pre-route Delay Prediction and Timing Optimization[4,5]

- Motivation … Improve prediction of pre-route arc delay/slew by ML, then import into P&R flow for improving timing → design time reduction & performance improvement
- Accurate 2-step prediction of "R, C, congestion → arc delay, arc slew" … supervised learning using ANN and CNN
- Import to P&R tool chain (Synopsys, Cadence) for improving timing closure



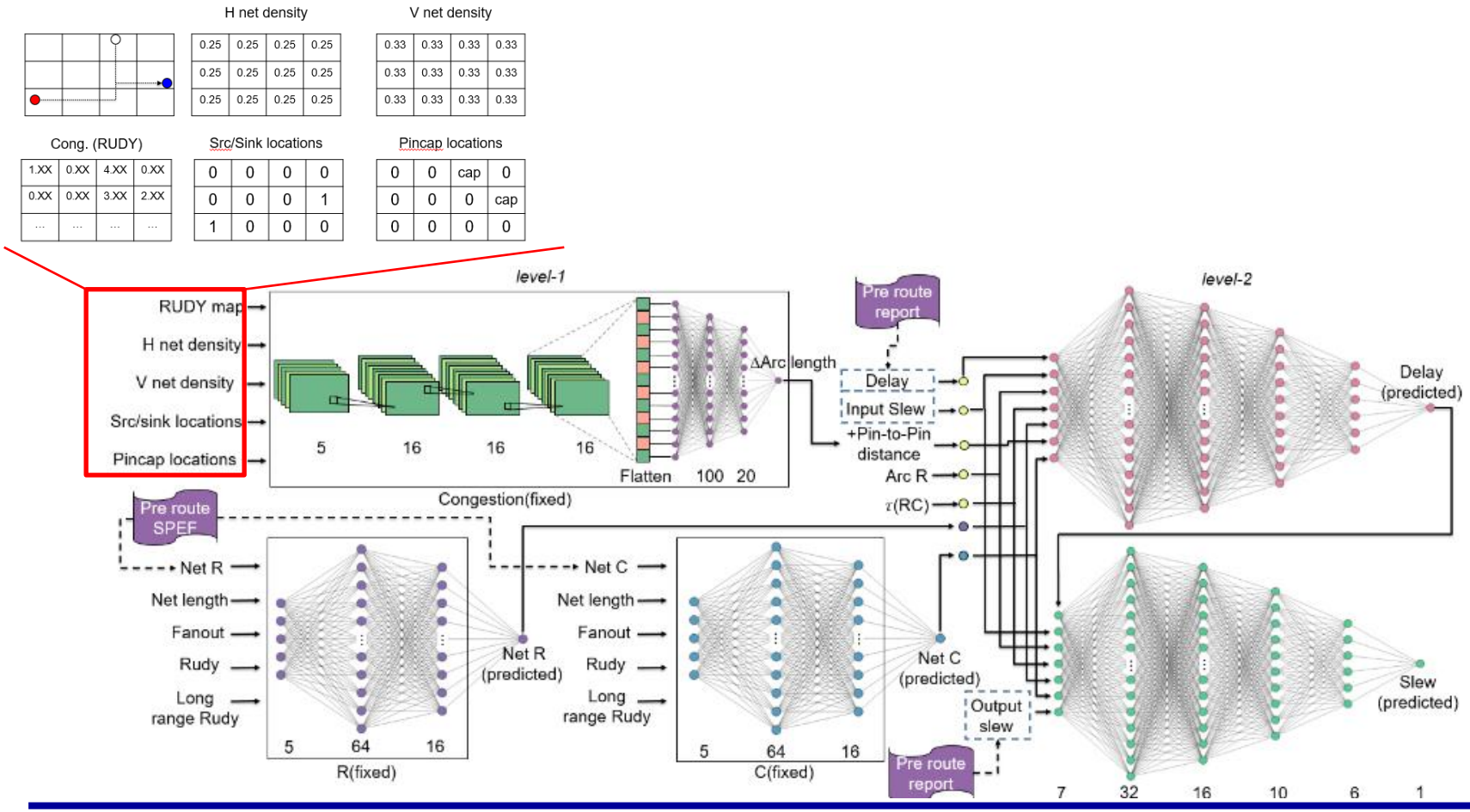
Evaluation dependency among pre-route timing parameters.



Flowchart of proposed framework.

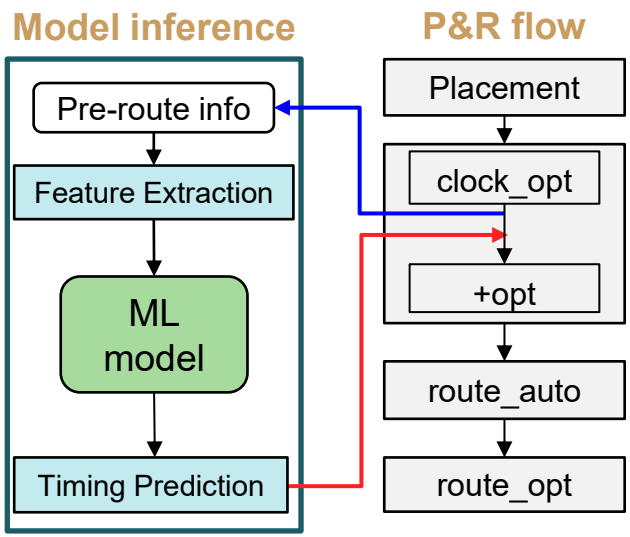
Pre-route Delay Prediction and Timing Optimization

- Delay Prediction - Levelized (Hierarchical) ML model
  - Sequential training process
  - Res. & Cap. & Cong. (ΔArc length) → Arc delay → Arc output slew



Pre-route Delay Prediction and Timing Optimization

- Timing Optimization
  - Pre-trained ML model → predict timing values for all arcs
    - Arc delay & output slew
  - Import timing values into commercial P&R tool
  - Run the tool's timing optimization engine
  - The tool will utilize accurate timing values



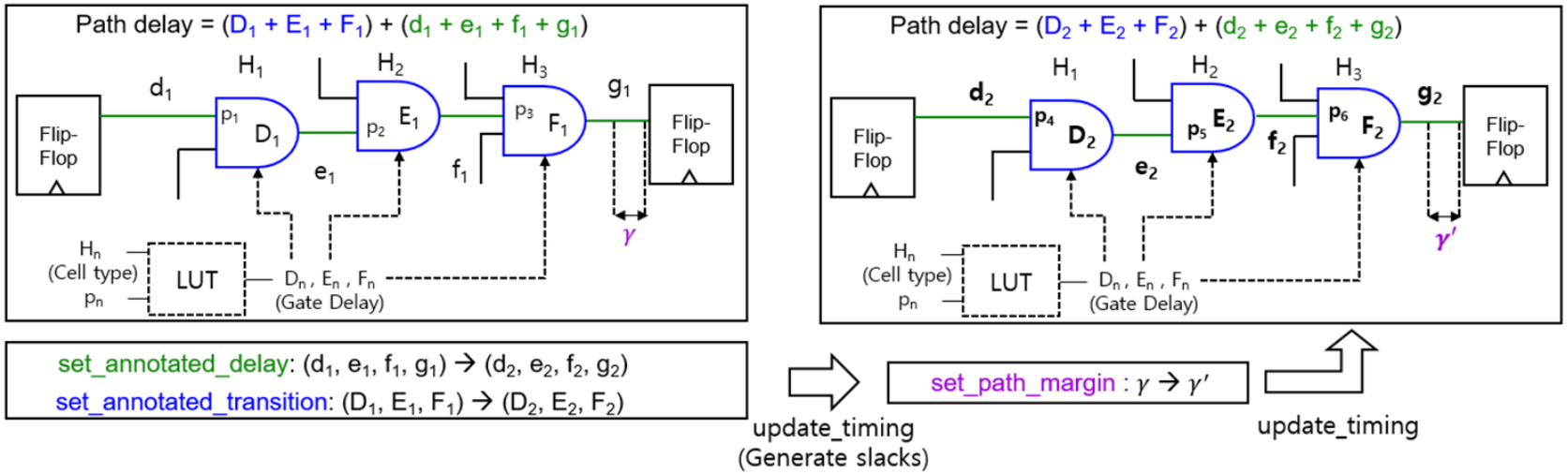
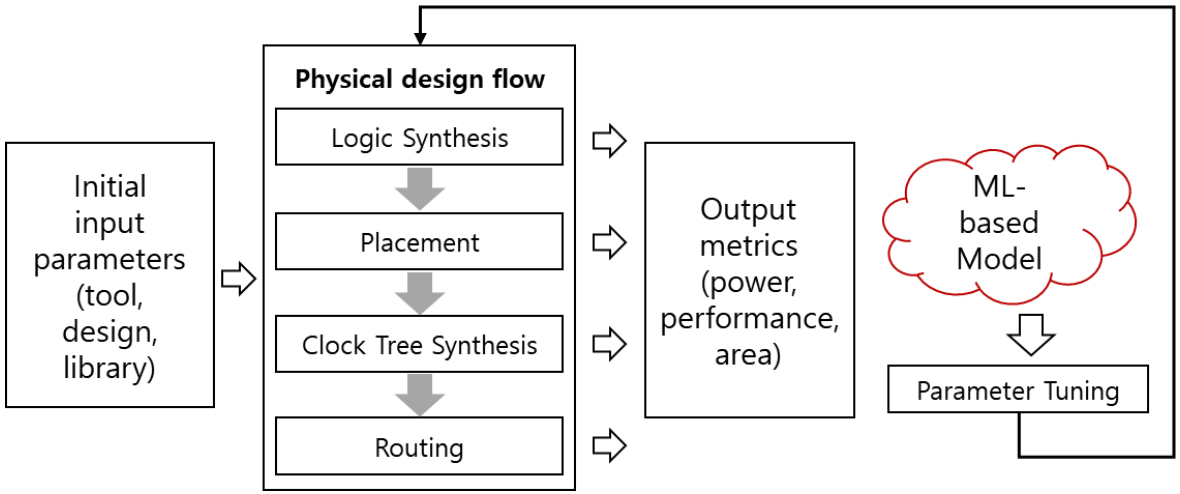


Figure: Overview of delay/slew import

Physical Design Optimization using DSO

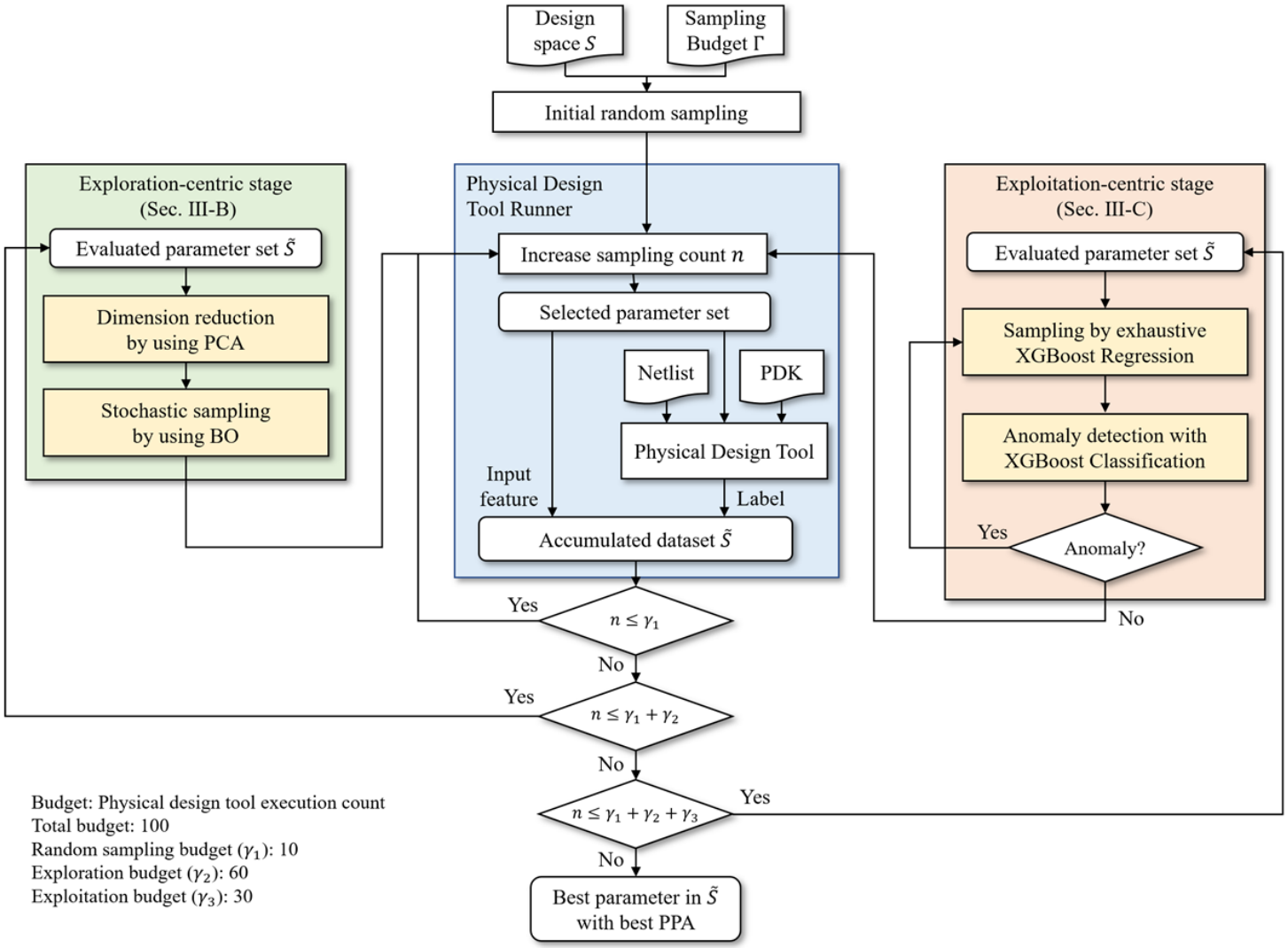
- Physical design(logic synthesis ~ P&R) optimization
  - Optimize output PPA by tuning input parameters through multiple stages of physical design flow
  - Impossible to get optimum solution in short time due to huge design exploration space
- Machine Learning(ML) based design space optimization(DSO) is required
- Developed physical design DSO framework (BOXGB) [6]
  - PCA assisted Bayesian Optimization + XGBoost augmentation with anomaly detection
- Developed multi-stage physical design DSO for PPA optimization



Ref: Input Parameters on Physical Design Stages

#	Parameter	Option	Stage			
			LS~ Place	Place-opt	CTS	Route
1	Max fanout	[8, 16, **24**, 32, 40, 48, 56, 64]				
2	Max transition	[10, 12.5, **15**, 17.5, 20] (%)				
3	Clock uncertainty	[**0**, 1, 2] (%)				
4	Global max density	[0.80, 0.85, 0.90, **0.95**]				
5	Clock power effort	[**low**, standard, high]				
6	Congestion effort	[low, **medium**, high]				
7	legalization inst gap	[**0**, 1, 2]				
8	uniform density	[true, **false**]				
9	wirelength opt level	[none, **medium**, high]				
10	Opt max density	[**0.80**, 0.85, 0.90, 0.95]				
11	reclaim area	[true, false, **default**]				
12	power effort	[**none**, low, high]				
13	early global route level	[standard / **medium** / low]				
14	CTS Buffer / Inverter Type	[**Option 1** / Option 2 / Option 3]				
15	CTS max fanout	[12, **20**, 28, 36, 44, 52]				
16	Skew target (%)	[10, **15**, 20] (%)				
17	CTS max transition (%)	[10, 12.5, **15**, 17.5, 20] (%)				
18	Route with timing driven	[true / **false**]				
19	Fixing DRC effort	[high / **medium** / low]				

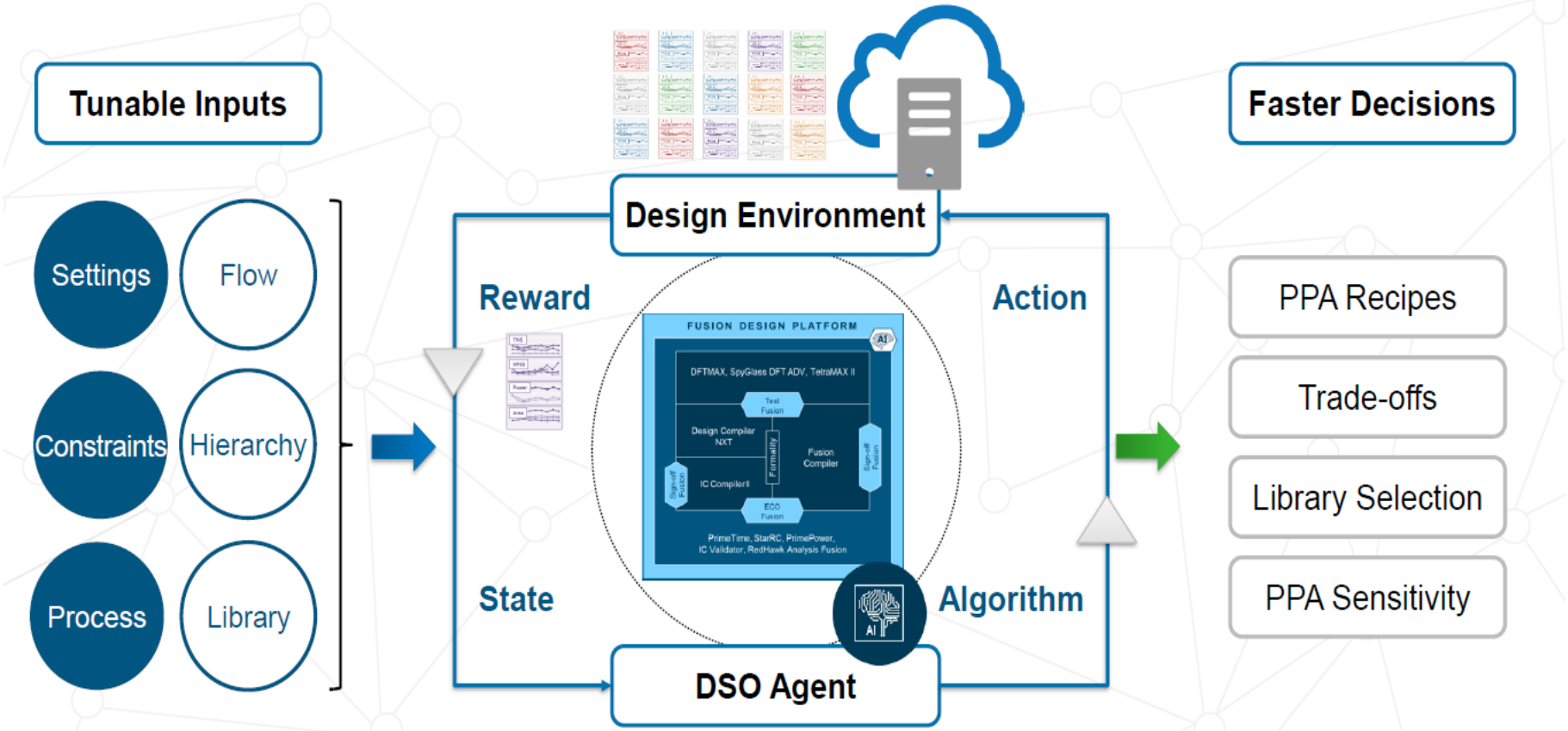
Physical Design DSO Framework – BOXGB [6]



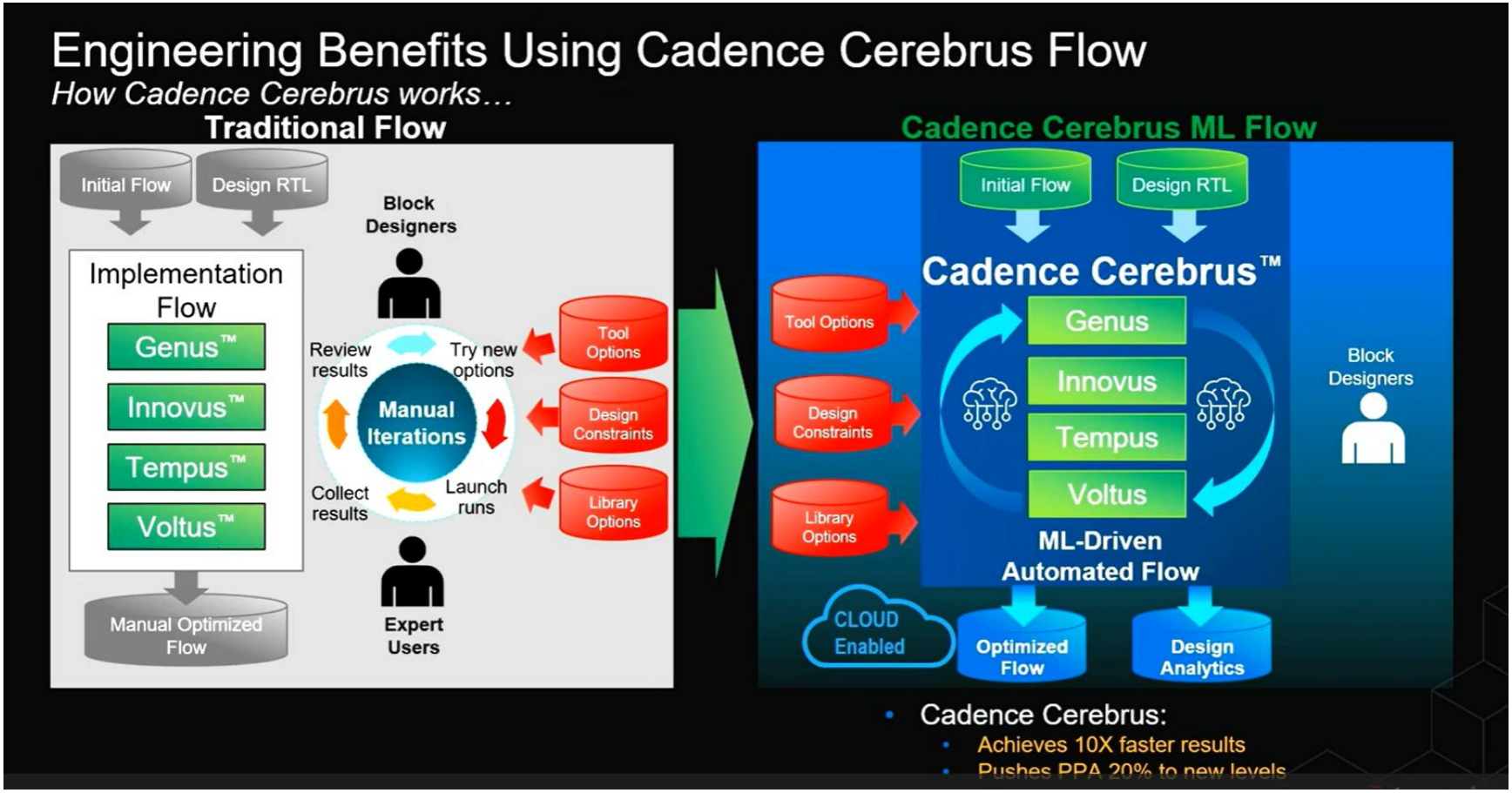
Physical Design Optimization using DSO - Synopsys

"DSO" App: AI-Driven Design Space Optimization for PPA

Uses reinforcement learning to navigate the technology-library-design solution space



Physical Design Optimization using DSO - Cadence



* Source: Michael Jackson, Cadence, "charting the path to 3nm and more than Moore" Tech Talk, DAC-2021

DTCO(Design Technology Co-Optimization)

- **DTCO** … Concurrent optimization of Technology(process/device) and Design
  - Determine TCAD/OPC model, PDK(spice/BEOL model, design rule), design infra(cell library, EDA tool, PDN architecture, metal stack) to produce block-level best PPA
  - Need several stages but closely related each other
  - Current DTCO work is time consuming and get only marginal effects due to insufficient space exploration and semi-manual preparation of inputs
  - **DTCO framework** → automated inputs generation, DSE/DSO, simulation/analysis speed-up
- **DTCO scope**

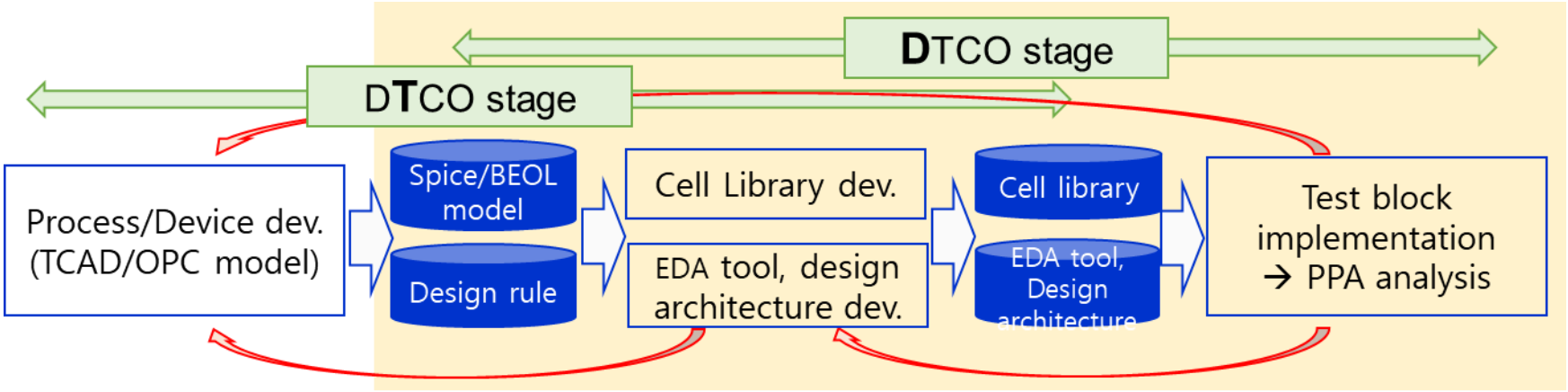


Figure: Overall flow of DTCO

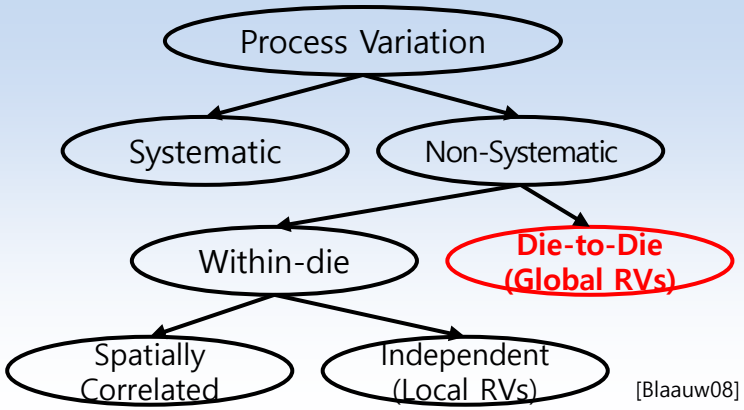
DTCO Framework

- **DTCO framework** … A system which enables the optimization of technology, design rule(DR), standard cell(SC) architecture by PPA analysis of digital block design
  - SC layout generator + SC characterization
  - DTCO framework
  - PDN and metal stack optimization
- **DTCO items**
  - FET variation … FinFET, NSFET, CFET, RVth/LVth/SLVth
  - SC architecture variation … #RT, Horizontal track pitch, CPP, single/multi-row, MPO, PWR width, cross-coupled tr., 1D/2D in-cell routing, DDB/SDB, w/wo COAG, CA connection, gear ratio, etc.
  - DR variation … minimum area rule, end-of-line rule, via rule
  - PPAC analysis using digital block design … area, performance, power

Variability aware DM

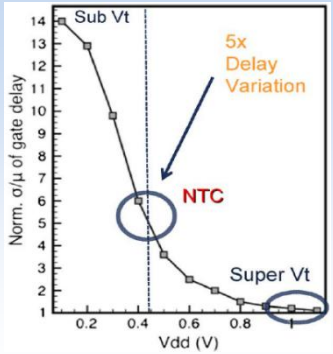
- **Introduction: Why HPM?**

Taxonomy of Process Variations

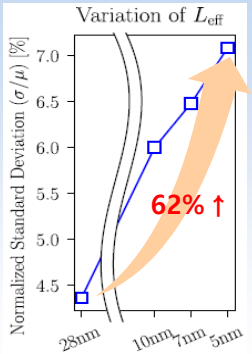


[Blaauw08]

Large Global Random Variation



[Dreslinski10]



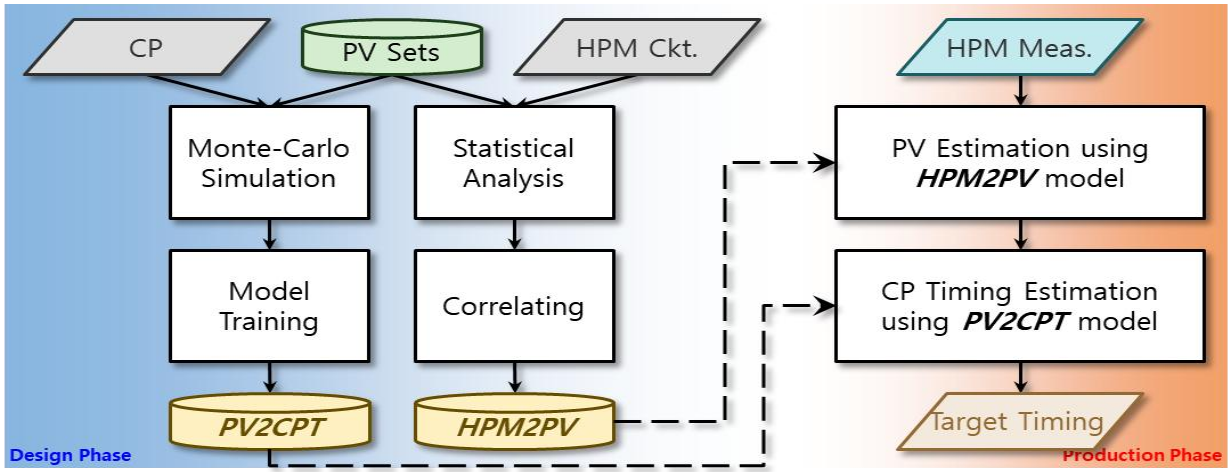
[IRDS17] and SEC 28nm PDK

– **Handling the variation for NTC and advanced process nodes**

Accurate & efficient HPM methodology is essential !

Variability aware DM

- **Proposed approach … Employing prior knowledge of PVs for building ML models** [9,10]

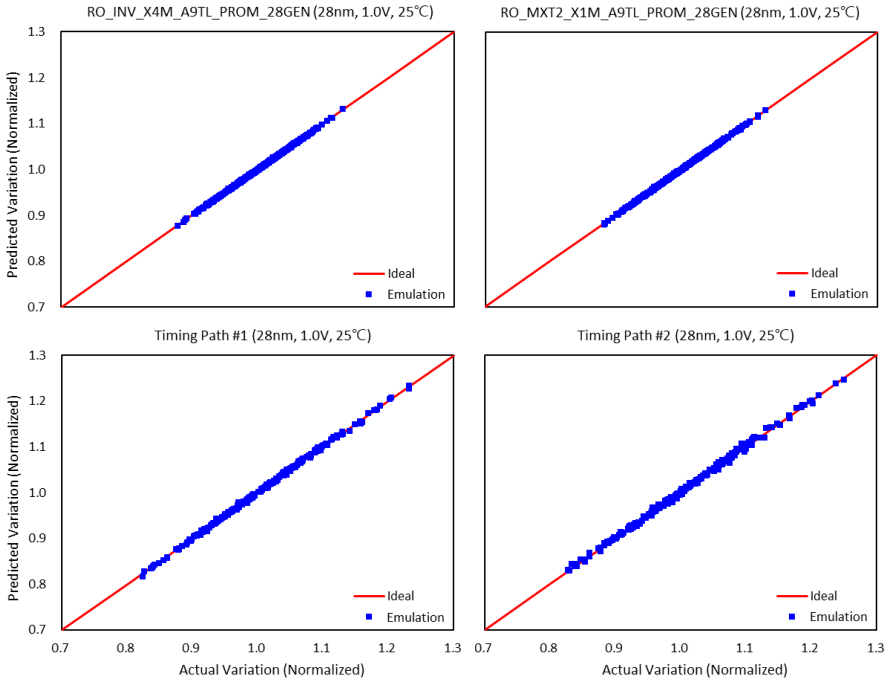


  - Two-step prediction flow considering design dependency
    - HPM2PV : Estimating PVs on each die from HPM measurements statistically
    - PV2CPT : Estimating CP timing from PVs using trained ML model

      → **ML approach: ANN model training and inferencing**
  - Statistical analysis and correlating … Optimization of HPM itself
  - Sophisticated ML model … Nonlinearity in NTC

Variability aware DM

- **ML approach: Surrogate model preparation**
  - Our methodology needs lots of SPICE simulations
    - Sensitivity analysis of Global PVs
    - Investigation of the impact of Local PVs
    - Preparation of training datasets for ML model
    - → Construct surrogate model instead of expensive SPICE simulations
  - Uncertainty quantification with tensor recovery *
  - Surrogate models for 28nm (FEOL, BEOL)



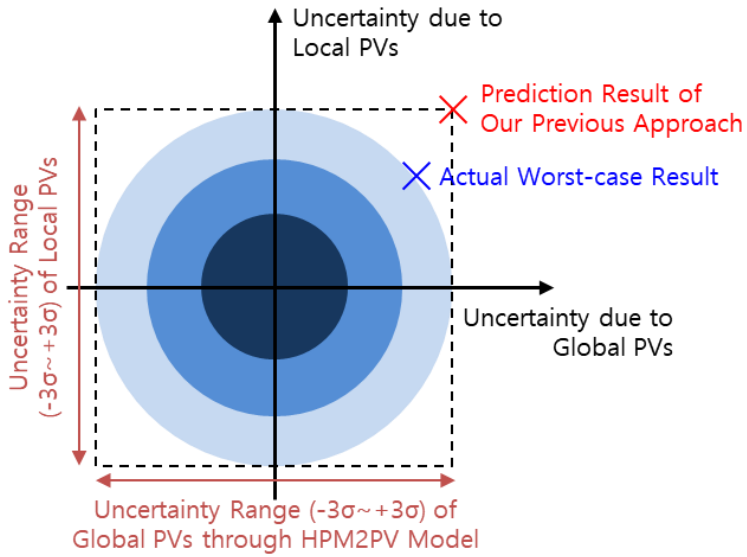
500 training samples / 250 validation samples / Total 19+20 parameters
Average and standard deviation of relative errors (on average)
- RO surrogate models: -0.0333% / 0.0852%
- Timing path surrogate models: -0.0201% / 0.255%

* reference: Z. Zhang et al., "Big-data tensor recovery for high-dimensional uncertainty quantification of process variations", IEEE Transaction on Components, Packaging and Manufacturing Technology, 2017.

Variability aware DM

- **ML approach: Uncertainty learning for HPM methodology**
  - **Uncertainty in deep learning**
    - Example of model uncertainty

    - Situations lead to uncertainty
      … Noisy data, Model uncertainty



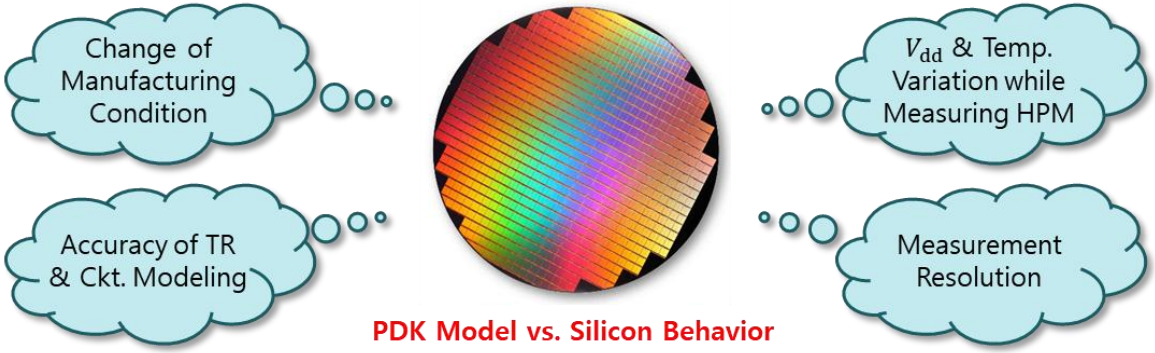
  - **Applying uncertainty learning to our PV2CPT model ***
    - Handling global & local PVs separately → Excessive pessimism insertion
    - Explicit addition of margin makes subsequent calibration difficult

    - Explicit global and local pessimism
      → Integrated & implicit pessimism (global + local)

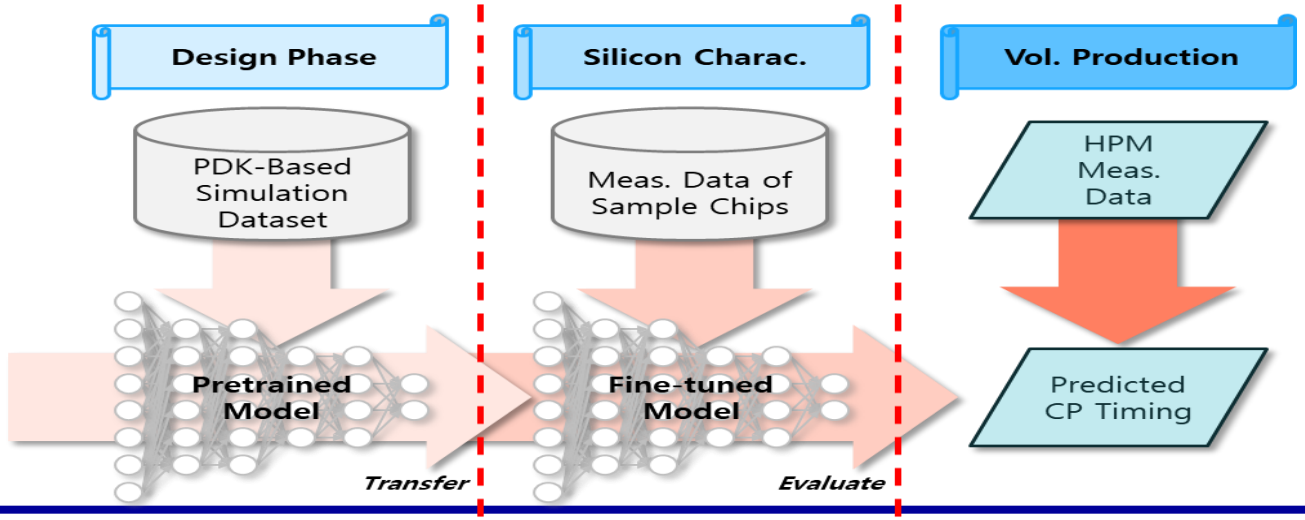
* reference: B. Lakshminarayanan et al., "Simple and scalable predictive uncertainty estimation using deep ensembles," Advances in Neural Information Processing Systems, 2017.

Variability aware DM

- **ML approach: Transfer learning for calibration**
  - **Necessity of calibrating prediction model**



  - **Transfer learning flow in our HPM methodology ***



* reference: S. Kang, "On effectiveness of transfer learning approach for neural network-based virtual metrology modeling," IEEE Transaction on Semiconductor Manufacturing, 2018.

Contents

- Technology Necessity and Current Status
- Research Status at Academia
- **Development and Adoption Status at Industry**
- Generative AI for Chip Design

Status of Semiconductor Industry - Nvidia

EDA RESEARCH STRATEGY

Understand longer-term potential for GPUs and AI in core EDA algorithms

Example – DREAMPlace:

RePlAce: Electrostatics-based nonlinear global placement

+

PyTorch

+

NVIDIA Volta GPU

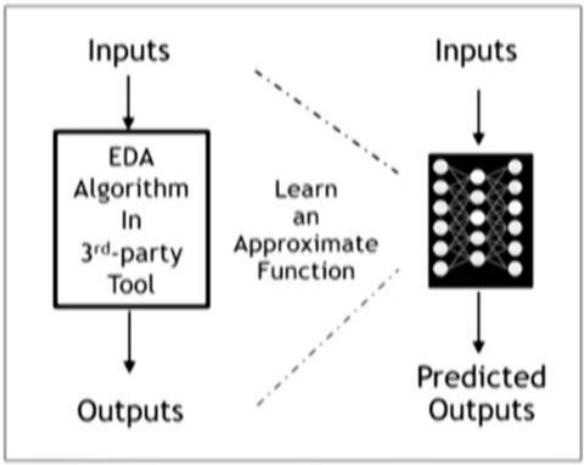
43x speedup on Global Placement

Accelerate known EDA algorithms with our AI Accelerated Computing platform

Use GPU programming systems and AI SW libraries for EDA

Potential Impact:

(1) Direct acceleration of known methods
(2) Scale up and scale out across a datacenter/cloud



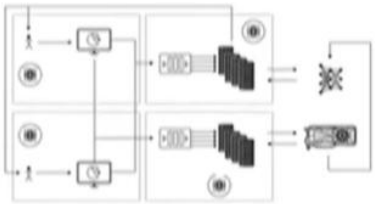
Trained ML models as predictors

Supervised learning with CNNs, GNNs, etc..
Applying latest NN models to EDA problems

Potential Impact:

(1) Augment traditional EDA w/ predictors
(2) Differentiable functions in optimizations

Current Work: Reinforcement Learning



Future Work: Generative Models

Develop new EDA algorithms using AI

Combine VLSI design/domain knowledge with AI methods for optimization

Potential Impact:

(1) Automate manual tasks
(2) Improve time-to-solution in EDA tools and flows with fewer iterations, faster steps

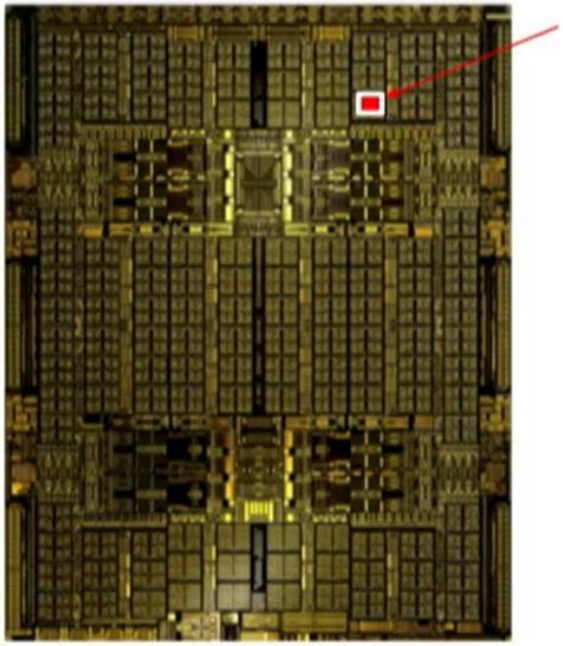
* Source: Bill Dally, Nvidia, "GPUs, Machine Learning, and EDA", Keynote at DAC 2021

Status of Semiconductor Industry - Nvidia

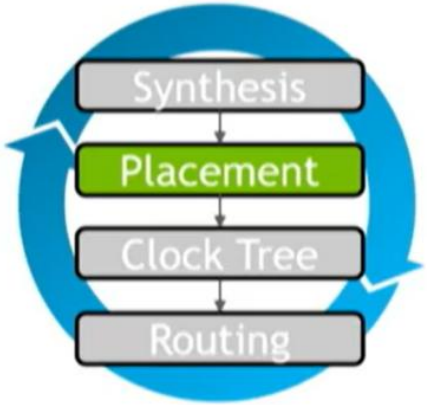
ACCELERATE PLACEMENT ON GPU WITH DL TOOLKITS

DREAMPlace (DAC 2019 best paper)
Collaboration between NVIDIA Research and UT-Austin

GA100 - Billions of standard cells



Large GPUs/SoCs divided into hundreds of unique *partitions*, with **millions** of cells per partition



Iteration time through backend flow
Days for one partition,
Many iterations per tapeout

How to improve scalability and dev productivity?

Use AI HW+SW platforms

PyTorch TensorFlow

NVIDIA CUDA

* Source: Bill Dally, Nvidia, "GPUs, Machine Learning, and EDA", Keynote at DAC 2021

Status of Semiconductor Industry - Nvidia

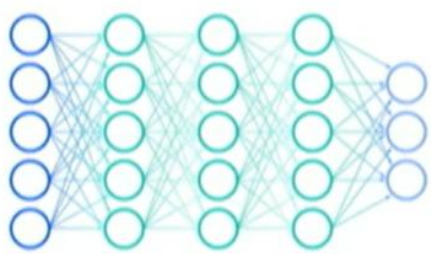
DEEP LEARNING ANALOGY

Deep Learning:
Optimize a differentiable loss function

$$\min_{w} \sum_{i}^{n} loss(\boldsymbol{f}(data_i, \boldsymbol{w}), label_i)$$

Forward compute $loss$

Batch of $data_i$, $label_i$



$loss$

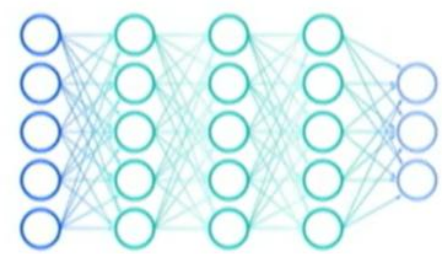
Backward compute $\frac{\partial loss}{\partial w}$

Non-Linear Global Placement:
Optimize a differentiable QoR function
TWL, density violation, ...

$$\min_{\boldsymbol{x},\boldsymbol{y}} QoR(\boldsymbol{x}, \boldsymbol{y})$$

Forward compute QoR

netlist



QoR

Backward compute $\frac{\partial QoR}{\partial x}, \frac{\partial QoR}{\partial y}$

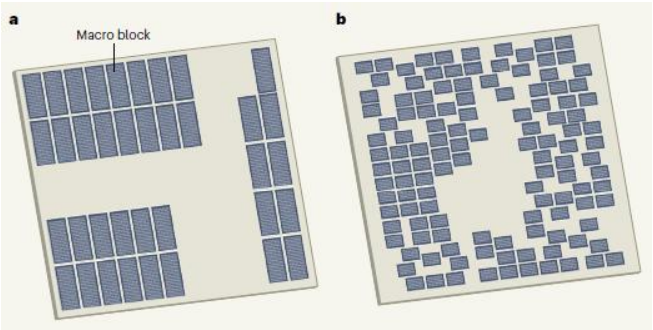
* Source: Bill Dally, Nvidia, "GPUs, Machine Learning, and EDA", Keynote at DAC 2021

Status of Semiconductor Industry - Google

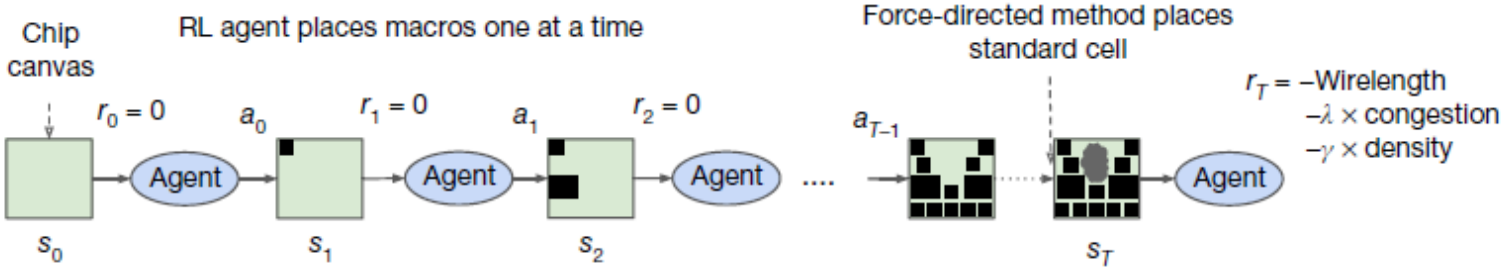
- Google presented at Nature ('21.6.9)
  - A deep RL(Reinforcement Learning) approach to chip floorplanning, using Edge-GNN architecture
  - In under 6 hours, it generates chip floorplans vs. several weeks of efforts by human experts
  - Used to design Google's TPU accelerator
  - State space of placing 1,000 clusters of nodes on a grid with 1,000 cells → $10^{2,500}$

    cf: state space of B/W stones in game Go → 10^{360}

"AI, 자신의 뇌를 직접 설계하다"



Microchip floorplans by
(a) humans and (b) Machine Learning



Overview of Google's Reinforcement Learning for chip floorplanning

* Source: Azalia Mirhoseini et.al., "A graph placement methodology for fast chip design", Nature, 10 June 2021

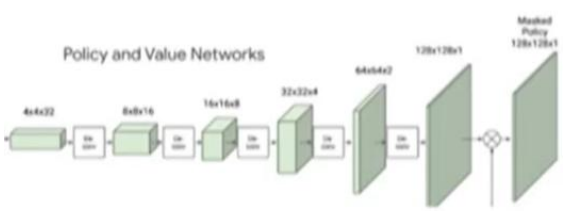
Status of Semiconductor Industry - Google

- Architecture exploration and RTL synthesis

Google Research

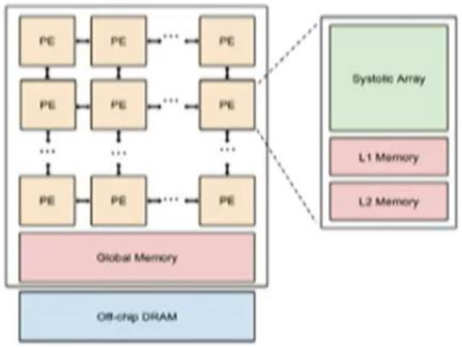
Using ML to Extend the Design Space Exploration

- Accelerator performance is a function of its hardware datapath and how workloads are mapped onto that datapath (by compilers or other software)
- Design space exploration typically only optimizes the datapath ($\sim 10^{10}$)
  - Can be a major limitation if a datapath variant has features which make current mapping or compilation approaches suboptimal
- **Simultaneously optimize** datapath, schedule, and compiler decisions
  - $O(10^{2300})$ search space




```
for b = [0:B):      // Batch
  for ih = [0:IH):    // Input Activation pixel (height)
    for iw = [0:IW):    // Input Activation pixel (width)
      for if = [0:IF):    // Input Activation pixel (channel)
        for kh = [0:KH):    // Kernel pixel (height)
          for kw = [0:KW):    // Kernel pixel (width)
            for of = [0:OF):    // Kernel number

              Output[b][ih][iw][of] +=
                Weight[kh][kw][if][of] * Input[b][ih+kh][iw+kw][if]
```



* Source: Jeff Dean, Google, "The potential of machine learning for hardware design", Keynote at DAC 2021

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- Technology Necessity and Current Status
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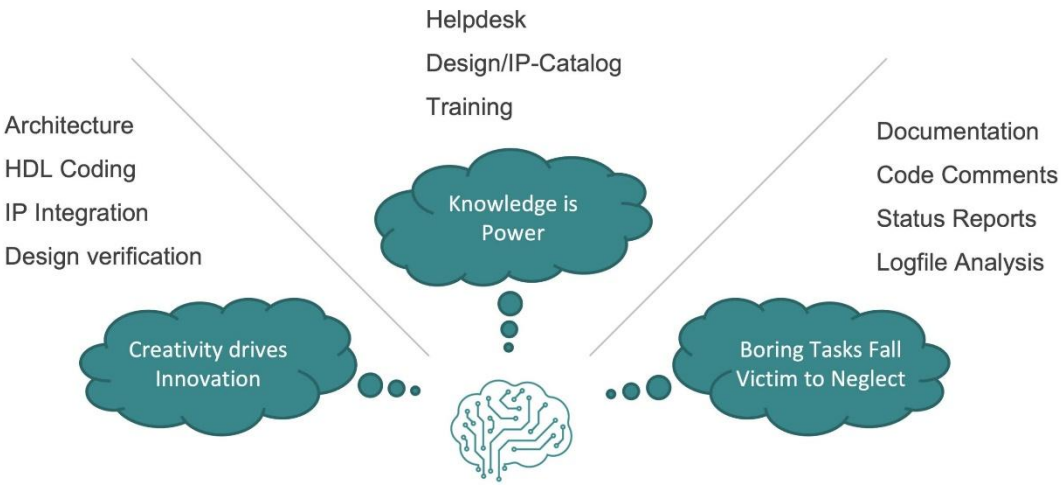
- **Generative AI for Chip Design**

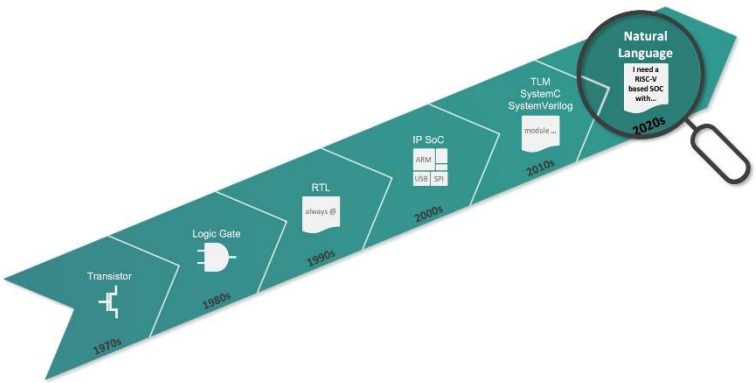
Generative AI for Chip Design

- Generative AI applications in chip design
  - Apply LLM for creation and innovation
  - Leverage LLM ability to help extract knowledge from information
  - Apply LLM in boring, mundane and repetitive tasks
- Design intent abstraction
  - Use natural language to specify design and verification intent

    → Translate a design intent expressed in natural language to HDL-code



Example application areas in chip design



Design intent abstraction over time

- source: Hans Bouwmeester, PrimisAI, "LLM and Chip Design", HotChips 2024, and "Fully automating chip design" @ https://semiwiki.com/eda/345291-fully-automating-chip-design/

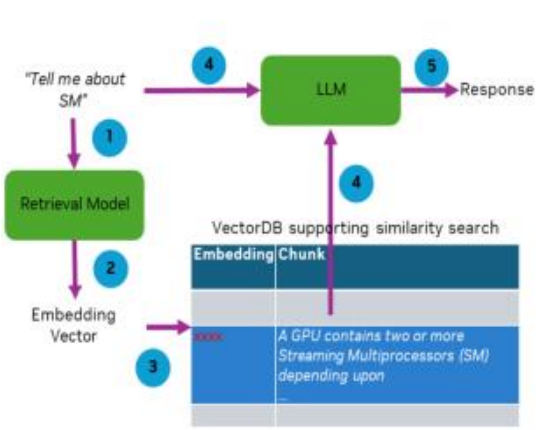
Generative AI for Chip Design – Current Status

- Status at EDA industries
  - Built "Synopsys.ai Copilot" framework (using Microsoft's Azure OpenAI service)
  - Built "Cadence.AI" framework
  - Support both of **Helpdesk and Documentation** mentioned previous page
- Status of applying **LLM for creation and innovation**
  - Recently, lots of researches are going on at universities and semiconductor industries (Nvidia etc.) … first **RTL HDL coding and validation**, but would be **extend more areas in entire chip design**
  - Now, quality are not enough for real adoption, and just starting stage, need to more development
  - In near future, enable to improve chip design productivity by providing design assistance as chatbots and copilots and automating more manual design tasks
  - Need a reliable and efficient inference infrastructure and more datasets and benchmarks

Generative AI for Chip Design – Nvidia

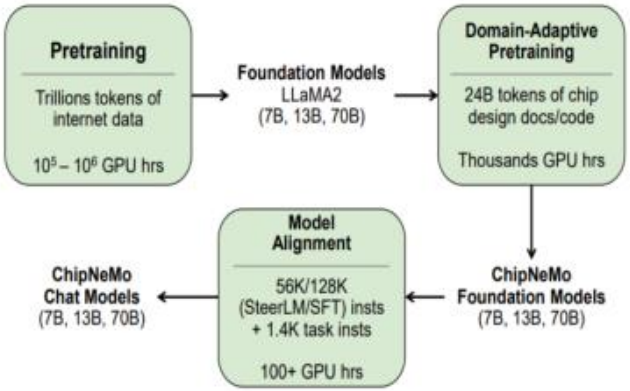
- How to make LLM learn to do chip design
  - **In-context learning** … LLM is augmented with a RAG knowledge-base
  - **Parameter training** and **Agent**

In-Context Learning



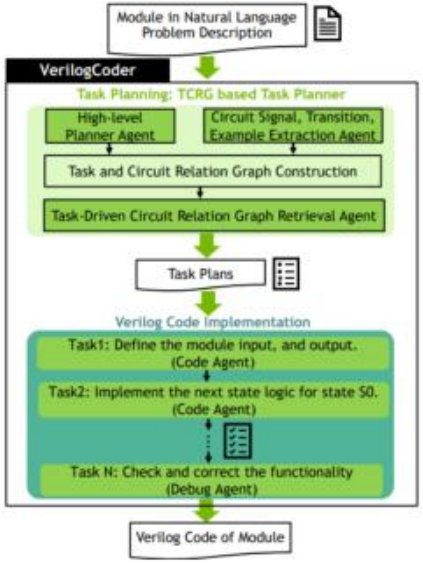
Retrieval Augmented Generation (RAG)

Parameter Training



M. Liu et al, ChipNeMo

Agent

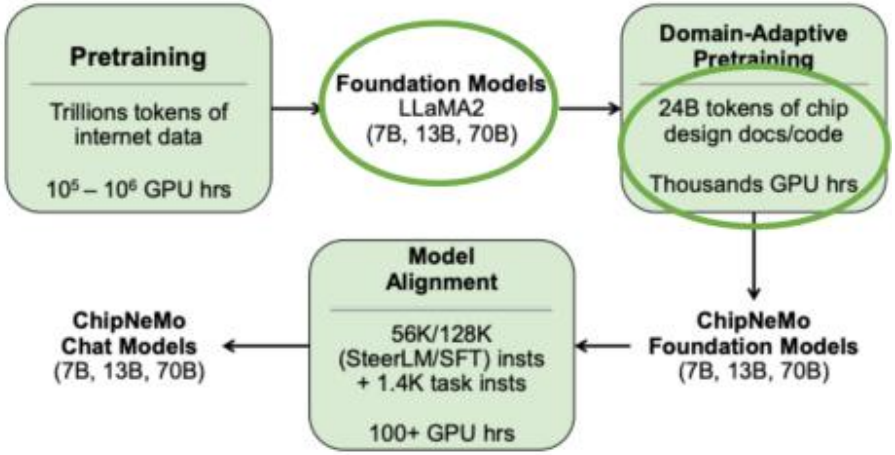


C.-T. Ho et al, VerilogCoder

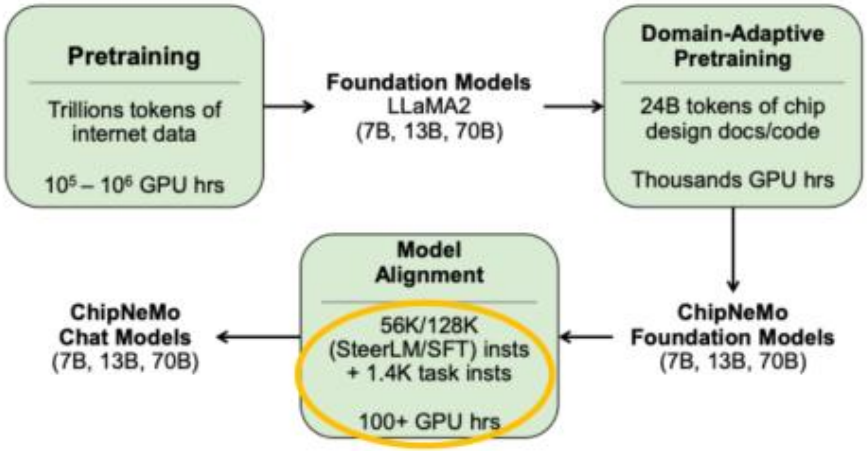
- source: Haoxong Ren, Nvidia, "Introduction to AI for chip design", HotChips 2024

Generative AI for Chip Design – Nvidia

- How to make LLM learn to do chip design → Parameter training → **Domain-adaptive LLM**
  - Model training stage → Domain-adaptive pre-training
  - Model fine-tuning stage → Task-specific fine-tuning
  - Model inference stage →
    - Prompt engineering … ChatBot models(ex: ChipNeMo), design prompts/queries
    - Agent … prompt engineering + simulator → feedback loop + self-correction
    - Retrieval-augmented generation … compiler logs



Domain-adaptive pre-training

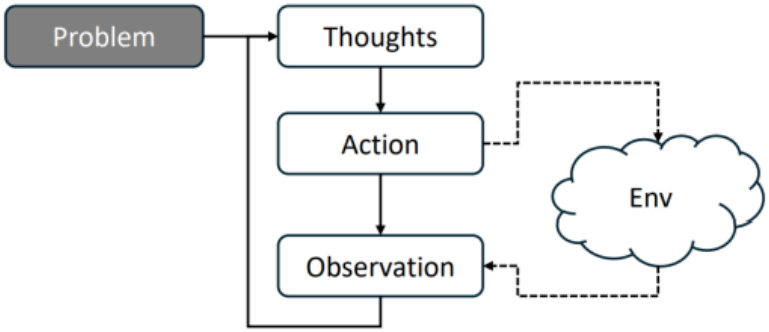


Task-specific fine-tuning

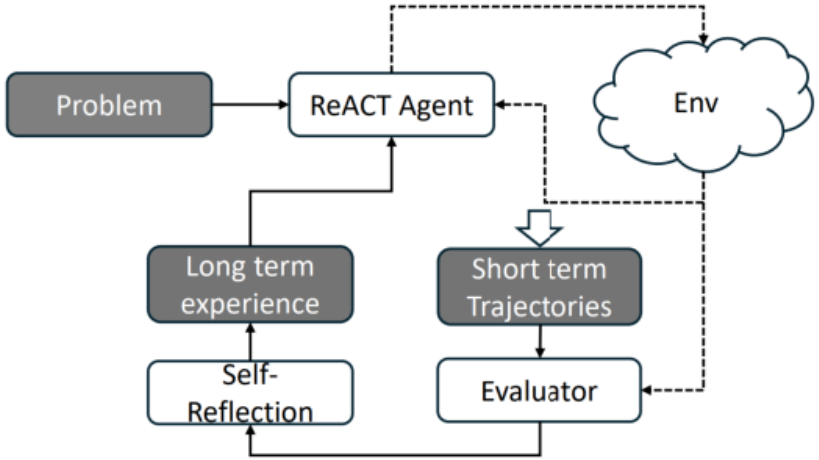
- source: Hanxian Huang, UCSD, "Domain-adaptive LLMs for chip design", HotChips 2024

Generative AI for Chip Design – Nvidia

- How to make LLM learn to do chip design → Agent
  - Non-agentic: simple question → LLM → answer
  - Agentic: complex question → LLM with (planning, memory, tools) → answer
    - Planning: without feedback, with feedback
      - ReACT: Thought, Action, Observation … Plan with environment feedback
      - Self-reflection: plan with feedback from accumulated experience
    - Memory: external files → archive, external vector DB → RAG
    - Tools: access tools for evaluation
  - Single agent and multi-agents



ReACT Agent



- source: Haoxong Ren, Nvidia, "LLM agents for chip design", HotChips 2024

Generative AI for Chip Design – Nvidia

- Agent example: VerilogCoder *

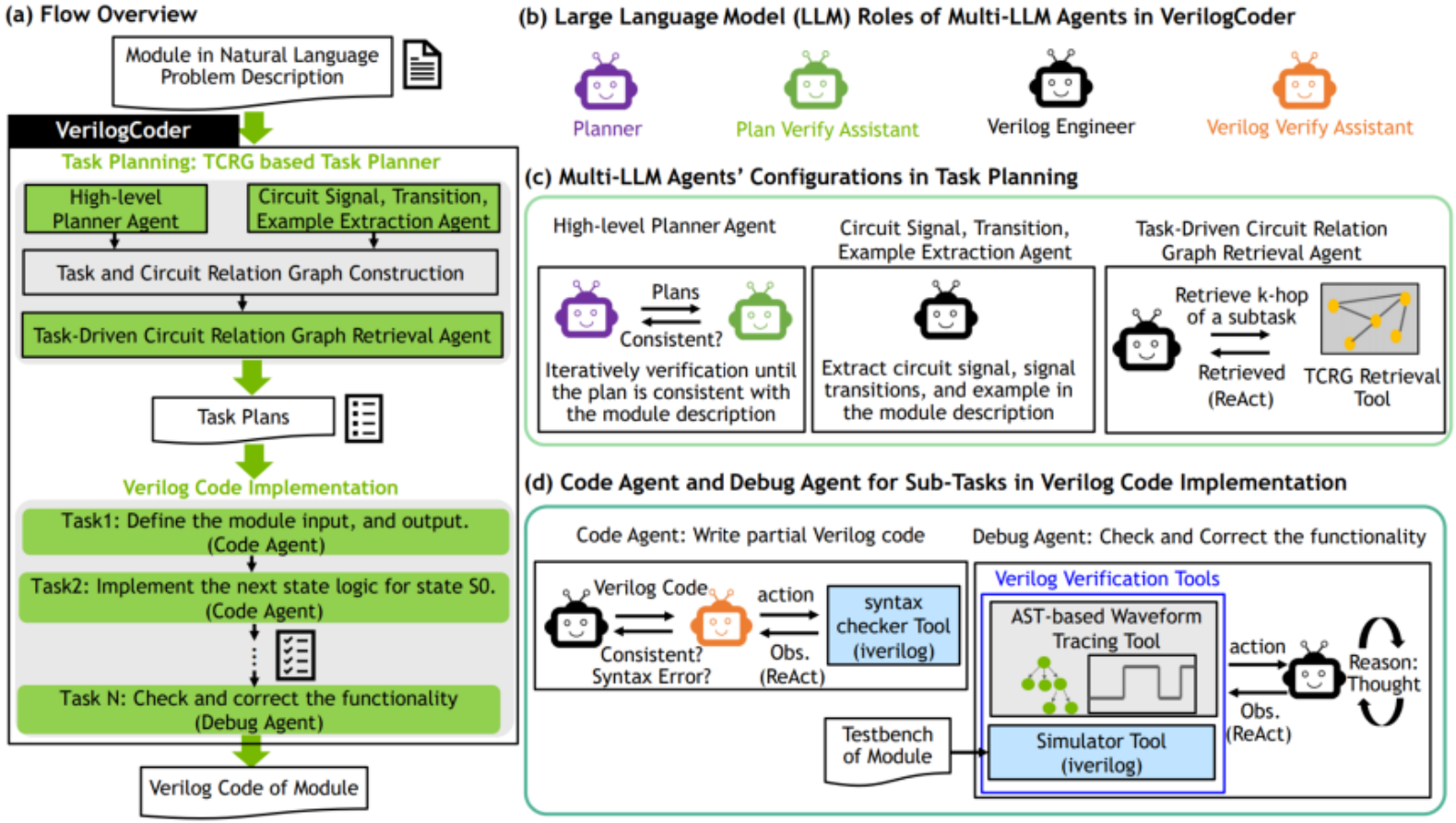


Figure 2: Flow overview of VerilogCoder. (a) Overall flow for Verilog code implementation task. (b) LLM roles of multi-LLM agents. (c) Multi-LLM agents in Task Planning. (d) Multi-LLM Agents for sub-tasks in Verilog Code Implementation.

* Chia-Tung Ho, et.al., "VerilogCoder: Autonomous Verilog Coding Agents with Graph-based Planning and Abstract Syntax Tree(AST)-based Waveform Tracing Tool", AAAI-2025

Generative AI for Chip Design – Nvidia

- Chip design assistance using LLM
  - Coding assistance … EDA script generation (generate scripts for specific tasks)
  - Know-how assistance … Engineering Chat Bot (answer questions about designs, infrastructures, tools, flows, HW domains, etc.)
  - Analysis assistance … Bug report analysis (summarizing bug report, predict task assignment)
  - ChipNeMo: M. Liu, et.al., "Domain-adapted LLMs for chip design"

EDA script

Question:
write a code in *TOOL1* to print the number of flop cells in the give rectangle (0,0,100,100)

Answer:

```
# Sure, here is a TOOL1 code to print the number of
↪   flop cells in the given rectangle
↪   (0,0,100,100):

# This code gets the top_ref of the design
top_ref = env().get_top_reference()
# This code gets the cells of the rectangle in
↪   top_ref
cells = env().layout().cells_in_rectangle( to
↪   rectangle(0, 0, 100, 100), hierarchical)
# This code gets the cells that are flop cells
flop_cells = cells.type("is_flop_cell")
# This code prints the number of flop cells
print(flop_cells.size())
```

Engineering Chat Bot

Question:
why do we add retiming stages?

Answer:
Retiming stages are added to improve the timing of the design and to account for the physical distance between different parts of the chip. By adding retiming stages, we can reduce the amount of time it takes for signals to travel between different parts of the chip, which can help to improve the overall performance of the design. Additionally, retiming stages can also help to reduce the amount of noise and interference that can occur on the signals, which can further improve the performance of the design.

Bug report

Prompt:
BugId : 002
Synopsis : Request power/current measurement on fabrics using fabric test module
Module : Silicon - DFT
Description :
Hi all,
We want to evaluate silicon current (power) consumed by fabric macros in a chip. This will help us correlate this power to pre-Si power estimates.
Here are the conditions we want to look at for this data:
Fabric test module activity: 0/25/50/75/100%
Corners:
typical_low_voltage
typical_high_voltage
We want to isolate fabric current (power) as much as possible, so we want to enable only the logic needed for fabric test modules and the fabrics

on can be broken down per module (there are 6 in the chi), that will help.
ormance
nal

ser A
ment only for current or for future chips as well ?
ser B
: be interest in future chips to perform this exercise as we push the boundaries of fabric power density for better area.
ser A
in this bug ?
ot clear.
specific needs with timelines and priority

nary:
ernal bug management system, a bug has been filed regarding the power and current measurements of the fabrics in the latest chip.
o evaluate the silicon current (power) consumed by the fabric macros in the latest chip and to correlate this power to pre-Si power
onditions for the measurements are the fabric test module activity at 0/25/50/75/100% and the corners are typical_low_voltage and
tage. The data needs to be broken down per module (there are 6 in the latest chip). Only the logic needed for fabric test modules
themselves need to be enabled.

nmary:
er B if the need for evaluating the silicon current of fabric macros in a chip is only for that specific chip or for future ones as
dies that there will likely be interest in future chips to perform this exercise as they push the boundaries of fabric power density for
: A then asks what is needed in this bug and asks for specific needs with timelines and priority.

ask Assignment:
User B

- source: Haoxong Ren, Nvidia, "Introduction to AI for chip design", HotChips 2024

Conclusion

- Status summary
  - Machine learning(ML) can be adopted to the diverse areas in semiconductor design and EDA technology, then will achieve the next level improvement for the QoR and design time.
  - ML will eventually bring EDA revolution, which is the old days' dream of the Silicon Compilation.
  - Integration of current EDA tools chain: "ML solution → current EDA tool chain + ML"
- Issues
  - Lots of researches vs initial stage for real adoption
  - Data preparation and training … Quantity & quality of data
    - Data ownership, security, collaboration issues between diverse groups (Foundries, Fabless semiconductor, EDA industries, universities)
  - ML solution maintenance
    - *Semiconductor technologies, models, designs, tools are changing continuously*
    - Need to build efficient data collection and maintenance methods
  - Need more collaboration between EDA companies, semiconductor companies (Foundries and Fabless design companies), universities

Reference

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3. HB Park, et.al., "Pin accessibility and routing congestion aware DRC hotspot prediction for designs in advanced technology nodes with consolidated practical applicability and sustainability", Transaction on CAD, 2024.
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7. HD Cho, et.al., "Standard cell layout generator amenable to design technology Co-optimization in advanced process nodes", DATE 2024.
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9. J. Heo, et.al., "Synthesis of Hardware performance monitoring and prediction flow adapting to near-threshold computing and advanced process nodes", ASPDAC 2020
10. J. Heo et.al., "Hardware Performance Monitoring Methodology at Near-Threshold Computing and Advanced Technology Nodes: from Design to Post-silicon", Transaction on CAD, 2021

Q & A

❖ 내용 관련 참고 자료 →

대한전자공학회 학회지

2024년12월호 특집: “AI반도체를 위한 HW/SW 기술” 中

“반도체설계를 위한 AI/ML 기술”